Cluster-by-time HDFE in did_multiplegt_dyn

Adding a focused option for FEs that are constant across groups within a time period

Worked example on data_test.csv: ac_uq_id #monthyear, a grid-cell panel where each grid (unique_small_grid_id) belongs to a single ac_uq_id.

1. The structure we want to absorb

> **Dataset (data_test.csv)**
> Panel at the (group, time) level:
> - unique_small_grid_id -> group id g
> - monthyear -> time id t
> - ac_uq_id -> coarser group attribute, maps each grid to a single value for all t (time-invariant at the group level)
>
> Goal. Absorb ac_uq_id #monthyear fixed effects -- one intercept per (cluster, time) cell -- alongside the time FE that did_multiplegt_dyn already partials out.

The defining feature: the absorbing variable is constant over groups inside the cluster and over the time periods we want to allow it to vary. Formally,

```
a_g = ac_uq_id_g  in {1,...,A}    (time-invariant)
absorb dimension:  (a_g, t)  in  {1,...,A} x {1,...,T}
```

This is the same structural property that trends_nonparam already exploits with one variable; the proposed option generalizes it to multiple such variables and routes the demeaning through reghdfe/hdfe.

2. Why this case deserves its own option

- The variance derivation in dCdH (2024) is built for the projection of Delta Y and Delta X_k onto time FE (optionally x a time-invariant stratum). Cluster x time FEs of the form a_g x t are a linear extension of that subspace; the algebra goes through verbatim.
- Section 1.4 of the companion paper already discusses trends_nonparam. We are not opening a new derivation -- we are saying 'allow multiple such strata, and use reghdfe as the implementation back-end.'
- Performance: one-way demean by (a_g, t, d) is the same bys ... gegen pattern already at L931-934. Two-way absorb (state#year industry#year) is what forces us to call reghdfe.
- Restricted scope keeps validation simple. The option does NOT accept slope FEs, time-varying categoricals, or unit FEs at a finer level than the group. Only 'time-invariant group attribute x time'.

3. Proposed syntax

```
did_multiplegt_dyn Y G T D, effects(L) [other options] ///
                  cluster_time_fe(varlist)
```

Semantics:

- Each variable a^(1), ..., a^(P) in the varlist must be time-invariant at the group level (checked at runtime).
- For each a^(p), the option adds the FE set {1{a^(p)_g = a0} . 1{t = t0}} to the projector used to residualize Delta Y and each Delta X_k, separately per baseline level d.
- Time FE and (optionally) trends_nonparam stay where they are; the new option just appends more dummies to F_d.
- If the user passes a variable that is not time-invariant at the group level, the command exits with a descriptive error.

Concrete call on data_test.csv

```
import delimited using "data_test.csv", clear
did_multiplegt_dyn Y unique_small_grid_id monthyear D, ///
    effects(4) placebo(2) controls(X1 X2)             ///
    cluster_time_fe(ac_uq_id)                         ///
    cluster(ac_uq_id)
```

4. Stage 1 -- the one stage that actually changes

Stage 1 partials a projector P_F_d out of Delta Y and each Delta X_k, then runs OLS on the residuals to recover theta_d. All we change is the projector.

Augmented FE space F_d

```
F_d = span{
        1{t = t0},                              // time FE (always)
        1{s_g = s0} . 1{t = t0},                // optional trends_nonparam
        1{a^(p)_g = a0} . 1{t = t0},  p=1..P    // cluster_time_fe (new)
      }, restricted to S_d (control sample).
```

The weighted projector P^W_{F_d} = Z (Z'WZ)^+ Z' W and its annihilator M^W_{F_d} = I - P^W_{F_d} project onto F_d in the weighted inner product W_d = diag(N_gt).

> **Bottom line**
> The MATH is identical to the existing Stage 1. Replace every appearance of 'demean by time FE within S_d' with 'demean by F_d within S_d'.

The four substitutions, line by line

- (1) Residualized FD of each control [L941]: resid_X_k = sqrt(N_gt) * [M^W_{F_d} . Delta X_k].
- (2) Outcome FD residual [L955]: diff_y_w = sqrt(N_gt) * [M^W_{F_d} . Delta Y]. (Old code did not partial time FE out of DY here; with multi-way absorb, FWL requires both sides to be demeaned by the same projector.)
- (3) Per-control score [L958]: prod_X_k = N_gt * [M^W_{F_d} . Delta X_k].
- (4) Fitted FE-only mean of DY [L984-996]: E_y_hat_gt^(d) = [P^W_{F_d} . Delta Y] (via reghdfe + residuals()).

What does NOT change in Stage 1

- matrix accum overall_XX = diff_y_wXX `mycontrols_XX' at L1018: same call.
- theta_d = M_d^{-1} q_d at L1034: same formula.
- Den_d^{-1} = M_d^{-1} . N_d^tot . G at L1052: same formula.
- Singular-cell guard at L1042-1090: same logic; just check rank(Z_d) = e(df_a) from reghdfe.

5. Stages 2 and 3 -- no math change, just plug in new theta_d

> **Zero code changes downstream**
> Stages 2 and 3 consume coefs_sq_`d'_XX, inv_Denom_`d'_XX, prod_X*_Ngt_XX and E_y_hat_gt_int_`d'_XX. Each is rebuilt from the augmented projector, so the downstream blocks at L4582-4689 (effects), L5145-5184 (placebos), and L4905-4954 (variance correction) automatically use the new HDFE structure. No line in those blocks needs to be touched.

For reference, the only points where Stage-1 outputs are read:

- L4679 -- Delta Y^(l) -= theta_d[k] . Delta X_k^(l).
- L5177 -- same line for the placebo long-difference.
- L4647, L4664 -- (Delta Y - E_y_hat_gt^(d)) inside the score in_sum_{k,d}.
- L4929 -- [Den_d^{-1} . in_sum_d]_{j,k}, the matrix-vector product in the variance correction.
- L4938 -- subtract theta_d[j] from the influence-function block.
- L4912-4946 -- prod_X_k inside the in_sum / U^var,X computation.

Every consumer is satisfied as long as the Stage-1 outputs reflect the augmented projection.

6. Stata patch using reghdfe / hdfe

Piece A -- Syntax and dependency check (L53, L66)

```
* L53 -- add the option:
syntax varlist(min=4 max=4 numeric) [if] [in] [, ///
    /* ... existing options ... */              ///
    cluster_time_fe(varlist)]

* After the gtools check at L65-73, add:
if "`cluster_time_fe'" != "" {
    qui cap which reghdfe
    if _rc {
        di as error "cluster_time_fe() requires reghdfe. Run: ssc install reghdfe"
        exit 198
    }
    qui cap which hdfe
    if _rc { di as error "... requires hdfe. Run: ssc install hdfe" ; exit 198 }

    * Each variable must be time-invariant at the group level.
    foreach v of varlist `cluster_time_fe' {
        tempvar sd_v
        bys `2': egen `sd_v' = sd(`v')
        sum `sd_v'
        if r(max) > 0 {
            di as error "cluster_time_fe(): `v' is not time-invariant at the group level."
            exit 198
        }
        drop `sd_v'
    }
}
```

Piece B -- Replace L919-961 (demean step)

```
* Build the FE set F_d to absorb.
local fe_set "time_XX"
if "`trends_nonparam'" != "" {
    capture drop trends_nonparam_temp_XX
    gegen trends_nonparam_temp_XX = group(`trends_nonparam')
    local fe_set "`fe_set' i.trends_nonparam_temp_XX#i.time_XX"
}
foreach v of varlist `cluster_time_fe' {
    local fe_set "`fe_set' i.`v'#i.time_XX"
}

local mycontrols_XX ""
local count_controls = 0
foreach var of varlist `controls' {
    local ++count_controls
    cap drop resid_X`count_controls'_time_FE_XX
    cap drop prod_X`count_controls'_Ngt_XX
    gen resid_X`count_controls'_time_FE_XX = .
    gen prod_X`count_controls'_Ngt_XX = .
    local mycontrols_XX "`mycontrols_XX' resid_X`count_controls'_time_FE_XX"
}
cap drop diff_y_wXX
gen diff_y_wXX = .

levelsof d_sq_int_XX, local(levels_d_sq_XX)
foreach l of local levels_d_sq_XX {
    tempvar smp
    gen byte `smp' = (ever_change_d_XX == 0       ///
                    & diff_y_XX != .              ///
                    & fd_X_all_non_missing_XX==1  ///
                    & d_sq_int_XX == `l'          ///
                    & time_XX < F_g_XX)

    local dvars "diff_y_XX"
    forvalues k = 1/`count_controls' { local dvars "`dvars' diff_X`k'_XX" }
    qui hdfe `dvars' [aw=N_gt_XX] if `smp', ///
        absorb(`fe_set') generate(_resXX_)

    replace diff_y_wXX = sqrt(N_gt_XX) * _resXX_diff_y_XX ///
        if d_sq_int_XX == `l' & `smp' == 1
    forvalues k = 1/`count_controls' {
        replace resid_X`k'_time_FE_XX = sqrt(N_gt_XX) * _resXX_diff_X`k'_XX ///
            if d_sq_int_XX == `l' & `smp' == 1
        replace resid_X`k'_time_FE_XX = 0 if missing(resid_X`k'_time_FE_XX) ///
            & d_sq_int_XX == `l'
        replace prod_X`k'_Ngt_XX = sqrt(N_gt_XX) * resid_X`k'_time_FE_XX ///
            if d_sq_int_XX == `l'
        replace prod_X`k'_Ngt_XX = 0 if missing(prod_X`k'_Ngt_XX) ///
            & d_sq_int_XX == `l'
    }
    drop _resXX_* `smp'
}
```

Piece C -- Replace L984-996 (FE-only fitted value)

```
foreach l of local levels_d_sq_XX {
    cap drop E_y_hat_gt_int_`l'_XX
    local controlsXX ""
    forvalues k = 1/`count_controls' { local controlsXX "`controlsXX' diff_X`k'_XX" }

    cap reghdfe diff_y_XX `controlsXX' [aw=N_gt_XX]    ///
        if d_sq_int_XX == `l' & time_XX < F_g_XX,      ///
        absorb(`fe_set') noconstant residuals(_res_DY_l_XX)
    scalar rc_XX = _rc

    if rc_XX == 0 {
        gen E_y_hat_gt_int_`l'_XX = diff_y_XX - _res_DY_l_XX ///
            if d_sq_int_XX == `l' & time_XX < F_g_XX
        drop _res_DY_l_XX
        scalar df_a_`l'_XX = e(df_a)
    }
    else {
        drop if d_sq_int_XX == `l'
        noi di "Baseline Treatment Level `l' dropped because of insufficient observations."
    }
}
```

The unchanged tail (L1018-1098)
matrix accum at L1018, theta_d = M_d^{-1} q_d at L1034, Den_d^{-1} = M_d^{-1} . N_d^tot . G at L1052, the singular-cell guard at L1042-1090, and the dropping of cells at L1094-1096 run verbatim on the new residuals. No code change.

7. End-to-end example on data_test.csv

```
* Path on the user's machine:
*   /Users/anzony.quisperojas/Documents/GitHub/did_multiplegt_dyn/Stata/data_test.csv

import delimited using "data_test.csv", clear

* Confirm the cluster-time FE candidate is time-invariant per group:
bys unique_small_grid_id: egen sd_ac = sd(ac_uq_id)
assert sd_ac == 0 | missing(sd_ac)
drop sd_ac

did_multiplegt_dyn Y unique_small_grid_id monthyear D, ///
    effects(4) placebo(2)                              ///
    controls(X1 X2)                                    ///
    cluster_time_fe(ac_uq_id)                          ///
    cluster(ac_uq_id)
```

What this estimates

For each baseline-treatment level d:

```
theta_d_hat = argmin_{theta, alpha^(d)}
    sum_{(g,t) in S_d} N_gt . [ Delta Y - sum_k theta_k Delta X_k - alpha^(d)_{a_g, t} ]^2

where alpha^(d)_{a, t} is an ac_uq_id-by-monthyear FE,
allowed to differ by baseline cell d.
```

The event-study estimators then use Y_{g,t} - Y_{g,t-l} - (Delta X^(l)_{g,t})' theta_d_hat as today, and the variance correction U^var,X uses the Den_d^{-1} built off this projection.

8. Sanity check: equivalence with trends_nonparam

Regression test before merging: passing a single a^(1) via cluster_time_fe(ac_uq_id) should give numerically identical event-study coefficients to the current command run with trends_nonparam(ac_uq_id), because F_d collapses to what trends_nonparam already builds at L991.

```
* OLD:
did_multiplegt_dyn Y unique_small_grid_id monthyear D,    ///
    effects(4) controls(X1 X2) trends_nonparam(ac_uq_id) ///
    cluster(ac_uq_id)
matrix b_old = e(b)
matrix V_old = e(V)

* NEW:
did_multiplegt_dyn Y unique_small_grid_id monthyear D,    ///
    effects(4) controls(X1 X2) cluster_time_fe(ac_uq_id) ///
    cluster(ac_uq_id)
matrix b_new = e(b)
matrix V_new = e(V)

matrix diff_b = b_old - b_new
matrix diff_V = V_old - V_new
matrix list diff_b
matrix list diff_V
```

Pass = max abs diff below ~1e-8. Extending to multi-FE (cluster_time_fe(ac_uq_id another_cluster)) has no trends_nonparam counterpart, but the FWL argument and the per-baseline-cell structure carry through.

9. Edge cases

- Group nested inside cluster. If unique_small_grid_id is finer than ac_uq_id, ac_uq_id is the cluster. If they are 1-to-1, ac_uq_id#monthyear reduces to (g,t) cells with one obs -- singularity guard fires.
- Singleton (cluster, year) cells. Use reghdfe's keepsingletons option; track e(num_singletons) for diagnostics.
- Missing values in cluster_time_fe variables. Drop the observation with a one-line warning.
- Interaction with trends_nonparam. The two options compose: F_d just stacks. reghdfe handles collinearity between a_g x t and a stratum s_g x t that overlaps with a_g via its rank-deficiency detector.
- Bootstrap and by_path(). Propagate cluster_time_fe(`cluster_time_fe') into the inner-call locals at L1337, L2030, L2033.

10. Summary

The proposal in one paragraph

Add an option cluster_time_fe(varlist) that takes time-invariant group attributes. Each is interacted with time and appended to the FE space F_d used to residualize Delta Y and Delta X_k inside the control sample for each baseline level d. The residualization is performed by hdfe (one call per baseline cell, demeaning Delta Y and every Delta X_k at once). The FE-only fitted value E_y_hat_gt_int_`d'_XX is rebuilt by reghdfe ..., absorb(...) residuals() and stored under the existing name. Nothing else changes: matrix accum, theta_d, Den_d^{-1}, the long-difference residualization at L4679, and the variance correction at L4905-4954 all consume the rebuilt Stage-1 objects and produce correct event-study estimates and variances by Frisch-Waugh-Lovell and the existing companion-paper derivation.